INCH-POUND

MIL-C-82890(NAVY)
9 August 1993

MILITARY SPECIFICATION

CARTRIDGE, FIRE EXTINGUISHER CCU-120/A

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the CCU-120/A Fire Extinguisher Cartridge, and is referred to herein as the "cartridge".

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-B-131	Barrier Materials, Watervaporproof, Greaseproof, Flexible, Heat-Sealable

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1377

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MIL-A-2550	Ammunition, General Specification for
MIL-C-10464	Cans, Hermetic Sealing, Metal, Light Gage, Tear-Strip Type
MIL-I-23659	Initiators, Electrical, General Design Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DOCUMENTS

INDIAN HEAD DIVISION, NAVAL SURFACE WARFARE CENTER (IHD/NSWC) (CAGE Code 14083)

504-174-TD-022	Detonator, Electric, BBU-43/A

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL1660AS100	Detonator, Electric, BBU-43/A
DL1660AS408	CCU-120/A Cartridge, Fire Extinguisher

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(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Data Review and Control Branch, (Code 8410P), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

PUBLICATIONS

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a first article sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Conformance to documents. The CCU-120/A Fire Extinguisher Cartridge covered by this specification shall be manufactured in accordance with DL1660AS408 and all documents listed thereon. The electric detonators shall be manufactured to conform to DL1660AS100 and Indian Head Division, Naval Surface Warfare Center (IHD/NSWC) 504-174-TD-022.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Bridge circuit resistance. The bridge circuit resistance of the cartridge, when measured in accordance with 4.6.2 shall be 0.42 to 0.75 ohms.

3.5 Environmental. The cartridge shall meet the environmental resistance requirements of 3.5.1 through 3.5.4.

3.5.1 Leakage. Each cartridge shall not exhibit a leak rate greater than 0.00001 (1×10^{-5}) cubic centimeters per second (cc/s) when tested in accordance with 4.6.3.

3.5.2 Vibration. The cartridge shall not autoignite, shall be safe to handle, show no evidence of disintegration of the propellant or derangement of components, and shall meet the requirements of 3.4, 3.6, 3.7, and 3.8, at 70°F following vibration testing of 4.6.4.

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3.5.3 Six-foot drop. The cartridge shall meet the functional requirements of 3.7 after the six-foot drop test (see 4.6.5).

3.5.4 Temperature-shock, humidity and altitude. The cartridge shall meet the requirements of 3.4 and 3.7 after being exposed to the temperature-shock, humidity and altitude environments specified in 4.6.6.

3.6 Radiographic examination. Radiographic inspection shall verify proper assembly, thread form, presence of parts, positioning of detonators, proper crimping and sealing and proper electrical connections. Film interpretation of the radiographs shall verify the presence and correct positioning of all parts and explosives (see 4.6.7).

3.7 Functional. The cartridge, when tested in accordance with 4.6.8, shall provide sufficient power to rupture the retaining disk of the container, permitting discharge of the fire extinguishing agent. There shall be no mechanical failure (see 6.5.1) or distortion that would impair removal of the cartridge.

3.7.1 Misfires. There shall be no misfires.

3.8 Workmanship. Workmanship shall be in accordance with the requirements specified on the applicable drawings, reference specifications, and 3.8.1 through 3.8.3.

3.8.1 Metal defects. The metal parts shall be free from cracks, splits, cold shunts, inclusions, porosity or any other metal defects.

3.8.2 Burr. The cartridge shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements,

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however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient (68 to 77°F)

b. Altitude: Sea level

c. Humidity: 50 percent relative, maximum

d. Pressure: Atmospheric (14.7 psia)

4.4 First article inspection. At the beginning of regular production, the contractor shall submit a first article sample to the Government approved facility for evaluation. The sample shall consist of 60 cartridges. Forty eight cartridges shall be expended in the tests as specified in Table I, and 12 cartridges retained for investigative purposes. The first article sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, excluding packaging and packing must be the same as used for regular production and shall be obtained from the same source of supply. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted.

4.4.1 Examination, tests and facility. The contractor shall inspect the sample in accordance with his inspection system. After provisional acceptance at the source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract or purchase order (see 6.2).

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table II.

4.5.1 Lot formation and identification. Unless otherwise specified in the contract or purchase order (see 6.2), formation and identification of lots shall be in accordance with MIL-STD-105 and MIL-STD-1168.

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TABLE I. First article inspection.

Test sequence	Test method	Quantity
Visual examination	4.6.1	60
Radiographic examination	4.6.7	60
Bridge circuit resistance test	4.6.2	60
Leakage test	4.6.3	60
Vibration test	4.6.4	12 (S/N 13-24)
Six-foot drop test	4.6.5	6 (S/N 25-30)
Temperature-shock, humidity, altitude (TSHA)	4.6.6	18 (S/N 31-48)
Radiographic examination	4.6.7	30 (S/N 19-48)
Bridge circuit resistance test	4.6.2	30 (S/N 19-48)
Functional: -65°F 70°F 200°F	4.6.8	 6 (S/N 01-06) 36 (S/N 13-48) 6 (S/N 07-12)

TABLE II. Quality conformance inspection.

Tests	Test method	Number of cartridges to be tested
Visual examination	4.6.1	All
Bridge circuit resistance test	4.6.2	All
Leakage test	4.6.3	All
Radiographic examination	4.6.7	All
Functional test: -65°F 70°F 200°F	4.6.8	1/3 at each temperature[1]

[1] Odd cartridges shall be conditioned at 200°F.

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4.5.2 Lot size. The production lot size shall be as specified in the contract or purchase order (see 6.2). The maximum lot size allowed is 500 exclusive of the test samples and shall consist of:

a. Components produced by a homogenous process.

b. Primer mixtures from not more than one lot.

c. Detonators all from the same lot.

d. Loading produced by a homogenous process.

4.6 Test methods and procedures. All tests shall be performed on cartridges which have been visually and dimensionally accepted. There shall be no dirt, oils, foreign matter, defective metals and the markings shall be clear, legible, and correct. Test and inspection methods and procedures are specified in 4.6.1 through 4.6.9.

4.6.1 Visual examination. One hundred percent examination shall be performed for all non-destructive critical defects. Examination for all major and minor defects shall be performed in accordance with the classification of defects. Four power (4X) magnification may be used for inspection of such defects as burrs and foreign matter. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All non-conforming items shall be rejected.

4.6.2 Bridge circuit resistance test. The bridge circuit resistance of the bridge circuit shall be measured from the terminal stud to the case using a test circuit which subjects the bridge circuit to a current of less than 50 milliamperes. This test shall be conducted in accordance with MIL-STD-202, Method 303. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection.

4.6.3 Leakage test. The leakage test shall be conducted using a leak detector of sufficient sensitivity to meet the requirements of 3.5.1. The test shall be a dry leak test at a pressure differential of 1.0 $\pm$ 0.1 atmospheres. Failure of any cartridge in the quality conformance test sample shall require 100% screening of the lot at the contractor's expense. Failure of any first article cartridge to comply with the requirements of 3.5.1 shall cause rejection of the first article.

4.6.4 Vibration test. The cartridge shall be subjected to vibration testing in accordance with MIL-STD-810B, Method 514.4, Category 4, Test Procedure I, except that each resonant and cycling period shall be divided equally at -65°F, 70°F, and 200°F. After vibration testing, the cartridges shall be checked radiographically and for bridge circuit resistance and meet the requirements of 3.5.2 before firing at 70°F. Failure of any cartridges to comply with the requirements of 3.5.2 shall be cause for rejection of the lot or first article.

4.6.5 Six-foot drop test. The first article sample shall be divided into three groups of 2 cartridges each. The six cartridges shall be dropped onto a 2-inch thick (minimum) embedded steel plate to impact as follows: Two with head up, two with head down, and two horizontal. No cartridge shall fire during this test. If the cartridges have successfully met the requirements of the

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six-foot drop test, they shall then be subjected to the tests as specified in 4.6.2, 4.6.3, and 4.6.7. The cartridges shall be fired at 70°F in accordance with 4.6.8 (see 3.5.3).

4.6.6 Temperature-shock, humidity, and altitude test. The temperature-shock, humidity, and altitude test shall be conducted in accordance with the requirements of MIL-I-23659. After environmental cycling, the cartridges shall meet the requirements of 3.4, 3.5.1, 3.6 and 3.7 at 70°F.

4.6.7 Radiographic examination. All assembled cartridges shall be radiographically examined in accordance with MIL-STD-453. Cartridges shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine with the dot on the flange up. Cartridges shall be identified with serial numbers prior to X-ray examination. Serial numbers shall be in consecutive order beginning with the number 001 in each lot. Any missing numbers are to be identified on the X-rays. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective cartridges found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot.

4.6.8 Functional test. The first article samples shall be ballistic tested as depicted in Table I. Quality conformance ballistic testing shall be conducted as depicted in Table II and the sample shall be randomly selected from the lot based on lot size in accordance with Table III. Sample cartridges shall be conditioned for a period not less than 4 hours nor more than 24 hours at the specified temperatures. The cartridges shall be removed from the conditioning chamber and test fired within 5 minutes. If any cartridge is not fired within 5 minutes after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The cartridges shall be initiated using a current of 8.0, +0.5, -0.0 amperes. The cartridge shall be fired in the cartridge test firing setup shown on Figure 1. A minimum of 8 ounces of water shall be added to the test fixture and the bottle (Walter Kidde P/N 899488 or equivalent) shall be pressurized to 600 +10 -0 pounds per square inch gauge (psig) prior to firing. The container has two discharge valve assemblies. Only one test cartridge shall be loaded into a discharge valve assembly per test shot. The other discharge valve assembly shall be sealed by a rupture disk. The cartridges shall meet the requirements specified in 3.7.

TABLE III. Ballistic sample size.

Lot size	Sample size	Samples to be retained
009 to 050	9	2
051 to 150	21	3
151 to 280	30	3
281 to 500	51	4

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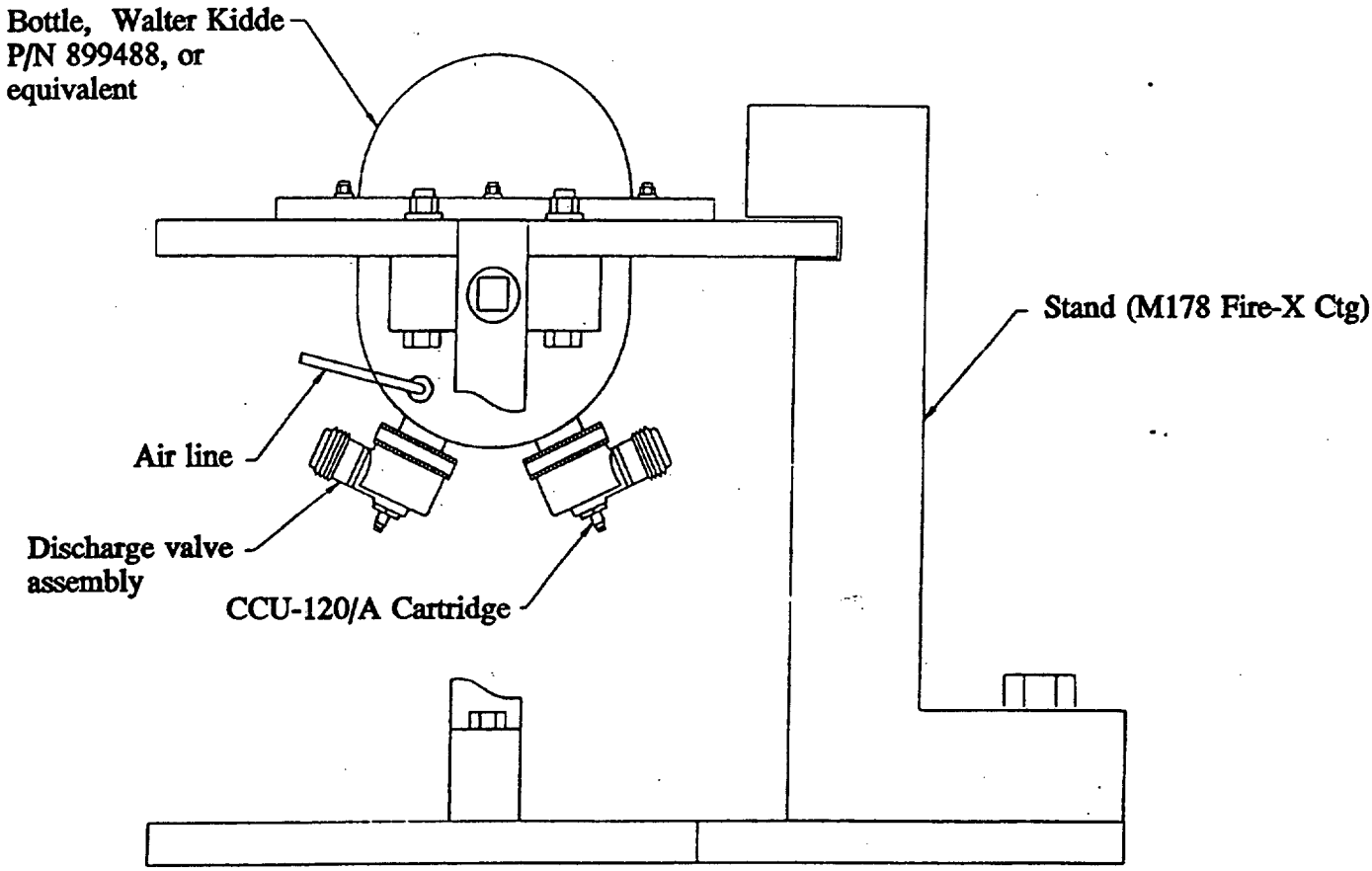


FIGURE 1. Cartridge test firing setup.

4.6.9 Inspection of packaging. The packaging, packing, and marking shall be visually examined for conformance to section 5.

5. PACKAGING

5.1 Packaging and packing. Level A packaging and packing shall be used for all production lots for service use. Level C packaging and packing shall be used for cartridges intended for expenditure in acceptance testing.

5.1.1 Level A.

5.1.1.1 Inner container. The inner container shall conform to MIL-C-10464, Type I. Each cartridge shall be packed with adequate dunnage and placed in a barrier conforming to MIL-B-131, Type I, Class 3. Two of these packs shall be placed in the can with adequate dunnage to provide a tight pack and the can hermetically sealed.

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5.1.1.2 Outer container. The number of outer containers as specified by the contracting activity, shall be packed in a nailed wood box in accordance with PPP-B-621, Class 2, Style 4 or 4½. As an alternate, a cleated plywood box in conforming to PPP-B-601, overseas type, Style A or B, may be used.

5.1.2 Level C. Level C packaging and packing shall be the manufacturer's best commercial practice to ensure arrival at the test facility without damage.

5.2 Marking. Lot numbers in accordance with MIL-STD-1168 shall be marked on the container. In addition to any special marking required by the contract or purchase order (see 6.2), all markings shall be in accordance with MIL-STD-129 and 49 CFR 171-178.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The cartridge is intended for use in releasing fire extinguishing liquid into the area surrounding an aircraft engine in the event of fire. This cartridge will be an additional cartridge available for the CH-53E and MH-53E fire extinguishing systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether a first article sample is required (see 3.1).

d. The facility designated to perform quality conformance inspection if other than the contractor (see 4.1).

e. Any special test conditions (see 4.3).

f. Test facility for first article inspection (see 4.4).

g. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.6.7).

h. Special marking, if required (see 5.2).

i. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this document is used as part of the description of

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work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.3 Hazard notice. The cartridge described herein and certain of it's components are flammable, toxic, or explosive and present hazards in manufacture, handling, storage and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environments, corrosive atmosphere, rough handling and electrically induced incidents.

6.4 First article. When a first article sample is required, the contracting officer should provide specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering such products which have been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definition.

6.5.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part other than a designed failure mode.

6.6 Subject term (key word) listing.

Electric detonation
Engine compartment
Fire bottle
Fire extinguisher container

Custodian:
Navy - OS

Preparing Activity:
Navy - OS
(Project 1377-0E47)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-82890	2. DOCUMENT DATE (YYMMDD) 93/08/09

3. DOCUMENT TITLE
CARTRIDGE, FIRE EXTINGUISHER CCU-120/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8420)	(301)743-4358/4510	354-4510/4358
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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